SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: MLA0303
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO3: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

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QUESTIONS: 10 **Total Marks: 50**

Annexure A

CODING CHALLENGE **Duration: 2hrs**

1.Predict the Reinforcement Learning (RL) framework by describing the agent–environment interaction, emphasizing the importance of states, actions, and rewards, and discuss how cumulative reward influences decision-making. (5 marks)

Answer:

Reinforcement Learning (RL) is a machine learning approach in which an agent learns to make decisions by interacting with an environment. The agent is not given the correct action directly; instead, it learns through rewards and penalties.

Agent–Environment Interaction

The interaction can be represented as:

State → Action → Reward → Next State → Action → ...

- Agent: Learner or decision-maker.
- Environment: The external system with which the agent interacts.
- State (S): Represents the current situation of the environment.
- Action (A): A decision selected by the agent.
- Reward (R): Numerical feedback received after an action.
- Policy (π): Strategy that determines which action to select.

For example, in a robot navigation system:

- State = robot's current position.
- Action = move left, right, forward, or backward.
- Reward = +10 for reaching the destination and -10 for hitting an obstacle.

Importance of State, Action and Reward

The state provides information required for decision-making. The action allows the agent to influence the environment. The reward tells the agent whether the selected action was good or bad.

The objective is not simply to maximize the immediate reward but to maximize the cumulative discounted reward:

where γ is the discount factor.

A high cumulative reward means that the agent has learned a sequence of actions that produces good long-term results.

Conclusion

Thus, RL learns through continuous agent–environment interaction. States describe situations, actions determine decisions, and rewards guide learning. Cumulative reward helps the agent prefer actions that provide better long-term outcomes.

2. Estimate how decision-making problems can be modeled as a Markov Decision Process (MDP), detailing the components of action space, transition probability, reward function, and discount factor. (5 marks)

Answer:

A **Markov Decision Process (MDP)** is a mathematical framework used to represent sequential decision-making problems where the result depends on the current state and selected action.

An MDP is represented as:

1. State Space (S)

The state space contains all possible situations of the environment.

Example: In a robot navigation problem, the robot's position and surrounding obstacles form the state.

2. Action Space (A)

The action space contains all actions that the agent can perform.

For example:

The available actions can be different for different states.

3. Transition Probability (P)

Transition probability describes the probability of reaching the next state after taking an action.

where:

- = current state
- = selected action
- = next state

For example, if a robot moves forward, there may be a 90% probability of reaching the intended location and a 10% probability of moving to another location due to uncertainty.

4. Reward Function (R)

The reward function provides immediate feedback:

Positive rewards encourage useful actions, while negative rewards discourage unwanted actions.

Example:

- Reaching destination = +100
- Hitting obstacle = -100
- Each movement = -1

5. Discount Factor (γ)

The discount factor determines the importance of future rewards.

- γ close to 0 → focuses on immediate rewards.
- γ close to 1 → considers future rewards strongly.

Markov Property

The important property of an MDP is:

The future depends on the current state and action, rather than the complete history.

Conclusion

MDPs provide a structured mathematical representation of decision-making problems. **State, action, transition probability, reward, and discount factor** together define the environment in which an RL agent learns an optimal policy.

3. Discuss the role of policy and value functions in RL, and examine how state-value and action-value functions contribute to evaluating long-term benefits of actions using the Bellman Equation. (5 marks)

Answer:

A **policy** is the strategy followed by an RL agent to select actions based on the current state.

It can be represented as:

which gives the probability of selecting action in state .

Policy

A deterministic policy directly selects an action:

A stochastic policy assigns probabilities to different actions.

For example, a robot may choose:

- Move forward = 70%
- Move left = 20%
- Move right = 10%

State-Value Function

The state-value function represents the expected cumulative reward obtained when starting from state and following policy π.

It answers:

"How good is this state in the long term?"

Action-Value Function

The action-value function or **Q-function** measures the expected cumulative reward for taking action in state and then following policy π.

It answers:

"How good is this action in this state?"

Bellman Equation

The Bellman equation expresses the value of the current state in terms of immediate reward and future value:

For Q-values:

Importance

Value functions help the agent evaluate long-term consequences. Instead of selecting an action only because it gives immediate reward, the agent considers the future reward as well.

Conclusion

The **policy determines what the agent does**, while **value functions evaluate how good states and actions are**. The Bellman equation connects immediate rewards with future rewards and forms the foundation of many RL algorithms.

4. Justify the exploration–exploitation trade-off in RL and compare strategies such as ε-greedy and softmax approaches in balancing learning efficiency. (5 marks)

Answer:

One of the major challenges in RL is deciding whether to:

- **Explore:** Try new actions to discover potentially better rewards.
- **Exploit:** Select the best action currently known.

If an agent only exploits, it may never discover a better strategy. If it only explores, it may waste time taking poor actions.

Therefore, a balance between exploration and exploitation is required.

ε-Greedy Strategy

In ε-greedy learning:

- With probability **ε**, choose a random action.
- With probability **1 − ε**, choose the action with the highest Q-value.

For example, if ε = 0.1:

- 10% of the time → explore.
- 90% of the time → exploit.

Advantages

- Very simple.
- Easy to implement.
- Computationally inexpensive.
- ε can be decreased gradually during training.

Limitations

- Random actions may be poor.
- It treats all non-greedy actions equally.

Softmax Strategy

Softmax selects actions according to their Q-values:

where τ is the **temperature parameter**.

- High τ → more exploration.
- Low τ → more exploitation.

Comparison

ε-Greedy	Softmax
Random exploration	Probability-based exploration
Simple	More refined
All non-best actions treated similarly	Better actions receive higher probability
Easy to implement	Requires temperature tuning

Conclusion

Both methods balance exploration and exploitation. **ε-greedy is simpler**, while **softmax provides more controlled exploration based on action quality**.

5. Explain how reward shaping can accelerate learning in RL and illustrate its impact on preventing suboptimal policies. (5 marks)

Answer:

Reward shaping is the process of modifying or adding rewards to provide additional guidance to an RL agent during learning.

In many RL problems, the final reward may be received only after completing a long sequence of actions. This creates a **sparse reward problem** and makes learning slow.

Example: Robot Navigation

Suppose a robot must reach a target.

Without reward shaping:

- Reach target → +100
- Otherwise → 0

The robot may need many trials before discovering the target.

With reward shaping:

- Reach target → +100
- Move closer → +5
- Move farther → -5
- Hit obstacle → -100
- Each unnecessary step → -1

The agent receives useful feedback at intermediate stages.

How Reward Shaping Accelerates Learning

Reward shaping:

1. Provides more frequent feedback.
2. Reduces the difficulty of sparse-reward problems.
3. Guides the agent toward useful states.
4. Reduces unnecessary exploration.
5. Helps the agent discover successful policies faster.

Preventing Suboptimal Policies

A poorly designed reward may cause an agent to learn an undesirable behavior.

For example, if a robot receives a reward simply for moving, it may continue moving without reaching its destination.

Therefore, reward design should align closely with the actual objective.

Reward Shaping and Policy

A suitable reward structure encourages the agent to learn:

This reduces the chance of getting stuck in a local or suboptimal strategy.

Limitation

Poor reward shaping can result in **reward hacking**, where the agent maximizes the artificial reward instead of accomplishing the real objective.

Conclusion

Reward shaping can significantly accelerate RL training and guide agents toward desirable behavior, but rewards must be carefully designed so that the learned policy matches the intended goal.

6. Identify the challenges of applying RL in real-world robotics and analyze how safety, sample efficiency, and environment variability affect agent performance. (5 marks)

Answer:

Applying reinforcement learning to real-world robotics is more difficult than applying it in simulation because physical environments are uncertain, expensive, and potentially dangerous.

1. Safety

RL requires exploration. In robotics, random exploration can cause:

- Collisions

- Hardware damage
- Injury
- Unsafe movements

Therefore, safety constraints and safe exploration methods are required.

2. Sample Efficiency

RL may require a very large number of interactions.

In simulation, millions of trials can be performed quickly. In real robots, each trial may require:

- Physical movement
- Energy
- Time
- Human supervision

Therefore, real-world RL can be expensive.

3. Environment Variability

Real environments continuously change.

Examples include:

- Lighting changes
- Moving objects
- Different surfaces
- Sensor noise
- Unexpected obstacles
- Weather conditions

An agent trained in one environment may perform poorly in another.

4. Sim-to-Real Gap

A policy trained in simulation may fail when transferred to a real robot because simulation cannot perfectly reproduce:

- Friction
- Sensor noise
- Mechanical errors
- Real-world physics

5. Hardware Limitations

Robot performance is also affected by battery life, computational resources, sensor accuracy, and mechanical constraints.

Effect on Agent Performance

Challenge	**Effect**
Safety	Limits exploration
Low sample efficiency	Slow and expensive learning
Environment variability	Reduces generalization
Sensor noise	Incorrect state estimation
Sim-to-real gap	Poor transfer

Challenge	Effect
Hardware limitations	Restricts actions

Conclusion

Successful real-world robotic RL requires safe exploration, sample-efficient algorithms, robust policies, simulation, transfer learning, and adaptation to changing environments.

7. Analyze the influence of the discount factor (γ) on short-term versus long-term decision-making, and evaluate its role in episodic tasks. (5 marks)

Answer:

The **discount factor γ** determines how strongly an RL agent values future rewards compared with immediate rewards.

The return is:

where:

Low Discount Factor

When:

the agent mainly considers immediate rewards.

Example:

A robot chooses an action that gives +10 immediately, even if another action gives +100 later.

Thus, low γ supports **short-term decision-making**.

High Discount Factor

When:

future rewards receive greater importance.

The agent may sacrifice immediate reward to achieve a better long-term outcome.

Example:

A robot may take a longer path now to reach a larger reward later.

Comparison

γ	**Behavior**
0	Only immediate reward
Low	Short-term focus
Medium	Balance
Close to 1	Long-term focus

Role in Episodic Tasks

An episodic task has a definite beginning and ending.

Examples:

- Playing a game
- Completing a maze
- Robot reaching a destination

A high γ is useful when success depends on a sequence of long-term decisions.

A lower γ may be useful when immediate rewards are more important or when future outcomes are highly uncertain.

Conclusion

The discount factor controls the **time horizon of decision-making**. Low γ encourages short-term behavior, while high γ encourages long-term planning. Selecting a suitable γ is important for achieving good performance in episodic RL tasks.

8. Evaluate the effectiveness of model-free RL compared to model-based RL in dynamic environments, and differentiate their strengths and limitations. (5 marks)

Answer:

Reinforcement learning approaches can broadly be divided into model-free and model-based methods.

Model-Free RL

Model-free RL does not explicitly learn the environment's transition model. Instead, it directly learns:

- A policy, or
- Value/Q-functions.

Examples include:

- Q-learning
- SARSA
- Policy Gradient

Advantages

1. Simple conceptual structure.
2. No requirement for an accurate environment model.

3. Can perform well in highly complex environments.

4. Useful when environment dynamics are difficult to model.

Limitations

1. Often requires many interactions.

2. Training can be computationally expensive.

3. Less effective when real-world samples are costly.

4. Usually has limited explicit planning capability.

Model-Based RL

Model-based RL uses a model of the environment:

and possibly the reward function to predict future states and rewards.

The agent can then plan before taking actions.

Advantages

1. Better sample efficiency.

2. Can perform planning.

3. Useful when real-world interaction is expensive.

4. Can adapt quickly when the learned model is accurate.

Limitations

1. Requires an accurate model.

2. Model learning can be computationally expensive.

3. Incorrect predictions can result in poor decisions.

4. More complex than model-free methods.

Comparison

Feature	**Model-Free**	**Model-Based**
Environment model	Not required	Required
Planning	Limited	Strong
Sample efficiency	Usually lower	Usually higher
Complexity	Lower	Higher
Model error	Not applicable	Can affect decisions
Examples	Q-learning, SARSA	Dyna-Q, planning-based RL

Dynamic Environments

In rapidly changing environments, model-free methods may be useful because they directly learn from experience. Model-based methods can be more efficient when the environment dynamics can be learned accurately.

Conclusion

Model-free RL is simpler and often robust to complex dynamics, while model-based RL is more sample-efficient and capable of planning. The appropriate choice depends on environmental complexity, availability of data, and cost of interaction.

9. Illustrate with an example how Q-learning updates the action-value function during training, and demonstrate its convergence behavior. (5 marks)

Answer:

Q-learning is a model-free, off-policy RL algorithm that learns the optimal action-value function:

The Q-learning update rule is:

where:

- = current Q-value
- α = learning rate
- r = immediate reward
- γ = discount factor
- = next state
- = best future Q-value

Numerical Example

Suppose:

Substitute into the equation:

Thus, the Q-value changes from **2 to 5.3** because the agent received a positive reward and expects future benefits.

Repeated Learning

As more experiences are collected, Q-values gradually become more accurate. Under suitable conditions such as sufficient exploration and appropriate learning rates, Q-learning converges toward the optimal Q-function.

Conclusion

Q-learning improves its estimates using immediate rewards and estimated future rewards. Repeated updates gradually reduce the difference between estimated and optimal Q-values, resulting in **convergence toward an optimal policy**.

10 Differentiate between on-policy and off-policy learning methods in RL, and assess their advantages and limitations in practical applications. (5 marks)

Answer:

RL algorithms can be classified based on whether the policy used for generating experience is the same policy being learned.

On-Policy Learning

An **on-policy algorithm learns about the same policy that it follows to interact with the environment**.

Example:

SARSA

The SARSA update is:

Here, the next action is the action actually selected by the current policy.

Advantages

- Learns from actual behavior.
- Naturally considers exploration.
- Can be safer when exploratory behavior affects the outcome.
- Useful in environments where the behavior policy itself matters.

Limitations

- Exploration can influence the learned policy.
- May converge to a more conservative policy.
- Usually less flexible than off-policy learning.

Off-Policy Learning

An **off-policy algorithm learns a target policy different from the policy used to generate experience**.

Example:

Q-learning

It uses:

The agent may explore using one behavior policy while learning the optimal greedy policy.

Advantages

- Can learn from exploratory experiences.

- Can reuse previously collected data.
- More flexible.
- Suitable for experience replay and offline learning.

Limitations

- Can be unstable in some complex settings.
- Poor or biased data can affect learning.
- Aggressive exploration can be unsafe in real-world applications.

Comparison

Feature	On-Policy	Off-Policy
Learning policy	Same as behavior policy	Different from behavior policy
Example	SARSA	Q-learning
Exploration	Directly affects learning	Separate from target policy
Flexibility	Lower	Higher
Data reuse	More limited	Better
Practical use	Safer/conservative behavior	Efficient learning from experience

Example

Suppose a robot has two actions:

- Safe route
- Risky route

An on-policy algorithm such as SARSA learns based on the actions it actually takes, including exploration. Q-learning can explore the risky route but still learn the optimal target policy based on the best estimated action.

Conclusion

On-policy methods learn the policy being followed, while off-policy methods can learn an optimal target policy from another behavior policy. On-policy learning can be preferable when actual behavior and safety are important, whereas off-policy learning is highly useful when experience reuse and learning efficiency are important.

Code of Conduct:

I **Ch. Venkata Lakshmi Hymavathi (192425397)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Ch. Hymavathi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately